

Technical Briefing: Characterization of Maximum Nash Welfare and EF1

Analysis of Fair Division for Indivisible Goods

1 Context and Definitions

In the allocation of indivisible goods, a **welfarist rule** selects an allocation maximizing a welfare function $f_n(u_1(A_1), \dots, u_n(A_n))$.

The **Maximum Nash Welfare (MNW)** rule specifically maximizes the product of agents' utilities. An allocation is **Envy-Free up to One Good (EF1)** if any envy between agent i and agent j can be eliminated by removing at most one good from agent j 's bundle.

Definition 1.1 (EF1). An allocation A is EF1 if for every pair of agents $i, j \in N$:

$$\exists g \in A_j \text{ such that } u_i(A_i) \geq u_i(A_j \setminus \{g\})$$

2 Theorem 3.3.1: Continuous Welfare Functions

Theorem 3.3.1 establishes that for a continuous and strictly increasing welfare function f_n , the rule guarantees EF1 if and only if it is equivalent to the MNW rule.

2.1 Lemma 3.3.2: Structure of the Welfare Function

This technical lemma proves that if f_n (continuous on $(0, \infty)^n$) satisfies a specific exchange symmetry, it must be a function of the product of its arguments.

- **The Condition:** $f_n((k+1)x_1, x_2, \dots, kx_i, \dots, x_n) = f_n(kx_1, x_2, \dots, (k+1)x_i, \dots, x_n)$ for all $k \in \mathbb{Z}^+$ and $i \in N \setminus \{1\}$.
- **Rational Density:** By defining $p(x) = f_n(x, z/x, \dots)$, the proof shows $p(rx) = p(x)$ for any positive rational r .
- **Continuity:** Because rational numbers are dense in $(0, \infty)$ and f_n is continuous, p must be constant.
- **Induction:** Backward induction demonstrates $f_n(x_1, \dots, x_n) = q(x_1 x_2 \dots x_n)$.

2.2 Proof Breakdown: Theorem 3.3.1 (b) $\Rightarrow$ (c)

- **Symmetry via Contradiction:** If Lemma 3.3.2 were false, a profile with $kn + 1$ goods could be constructed where a welfarist rule picks a non-EF1 allocation.
 - **Profile Construction:** Utilities are set to $u_j(g) = x_j$ or x_j/k , with a "bridge good" g_m having utility $x_1 - \epsilon$.
 - **Welfare Maximization:** If f_n is not the MNW rule, the rule favors an agent receiving $k + 1$ goods, violating EF1.
 - **Zero Utility:** f_n must favor positive products over zero-utility allocations to remain strictly increasing.
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3 Theorem 3.4.1: Non-Continuous Welfare Functions

Theorem 3.4.1 states that f_n guarantees EF1 if and only if:

$$f_n(x_1, \dots, x_n) > f_n(y_1, \dots, y_n) \iff \prod x_i > \prod y_i$$

3.1 Lemma 3.4.2: The Exchange Inequality

In a uniform profile where goods have utility x_i/k , any EF1 rule must satisfy:

$$f_n(x_1, \dots, x_n) \geq f_n\left(x_1, \dots, \left(1 + \frac{1}{k}\right)x_i, \dots, \left(1 - \frac{1}{k}\right)x_j, \dots, x_n\right)$$

3.2 Lemma 3.4.3: Comparison with Unit Utility

For $0 < z < x_1 x_i$, it holds that $f_n(x_1, \dots, x_i, \dots) > f_n(z, \dots, 1, \dots)$. This uses sequences $h_1(k) = (1 + 1/k)^{\lfloor k \log x_2 \rfloor} \rightarrow x_2$ and $h_2(k) = (1 - 1/k)^{\lfloor k \log x_2 \rfloor} \rightarrow 1/x_2$ to bridge utilities to the unit benchmark.

3.3 Lemma 3.4.4: Multi-Agent Generalization

For any $x_1, \dots, x_n$, if $z < \prod x_i < y$, then:

$$f_n(y, 1, \dots, 1) > f_n(x_1, \dots, x_n) > f_n(z, 1, \dots, 1)$$

3.4 Final Proof Synthesis

- **Case 1 (Positive):** If $\prod x_i > \prod y_i > 0$, we use a mean product w and Lemma 3.4.4 to show $f_n(\mathbf{x}) > f_n(w, \mathbf{1}) > f_n(\mathbf{y})$.
- **Case 2 (Zero):** If $\prod y_i = 0$, we compare y to a positive set z_i where $\prod z_i = X/2$.

4 Summary Table of Proof Steps

Result	Primary Function	Core Proof Technique
Lemma 3.3.2	Proves f_n depends on product.	Rational density and continuity.
Theorem 3.3.1	Characterizes continuous MNW.	Profile construction ($kn + 1$ goods).
Lemma 3.4.2	Basic exchange constraint.	Uniform utility profile analysis.
Lemma 3.4.3	Compares x_i to unit utility.	Squeeze theorem and repeated exchanges.
Lemma 3.4.4	Generalizes to n agents.	Iterative application of Lemma 3.4.3.
Theorem 3.4.1	Characterizes non-continuous MNW.	Mean-value comparison.

5 Key Conclusion

“MNW is the unique additive welfarist rule that guarantees EF1... the MNW rule is in fact the only (not necessarily additive) welfarist rule that guarantees EF1.” (Section 3.1)